

ESG Intelligence Briefing — September 26, 2026

Executive Summary

The EU's Carbon Border Adjustment Mechanism (CBAM) is now fully effective, imposing financial obligations on importers of carbon-intensive materials like steel, cement, and aluminium from Kazakhstan into the EU. Simultaneously, Kazakhstan's new Construction Code, effective July 1, 2026, introduces stricter controls, digital cadaster, and longer guarantee periods, demanding greater supply chain transparency. Construction material production in Kazakhstan grew 9.4% in H1 2026, but continued import reliance and CBAM exposure necessitate an urgent shift towards verified low-carbon materials and sophisticated supply chain management to mitigate costs and enhance competitive advantage.

What Changed Since Last Briefing

Previous briefings highlighted the impending impact of CBAM and the need for low-carbon materials, but quantitative analysis was hampered by tool errors. Today's update confirms CBAM's full operational status with financial implications from January 2026 and provides concrete details on Kazakhstan's new Construction Code, which commenced on July 1, 2026. While specific embodied carbon calculations remain challenging due to ongoing tool limitations, the focus on low-carbon innovations in concrete provides pathways for mitigation.

1. Regulation Watch

- **EU Carbon Border Adjustment Mechanism (CBAM) Fully Operational:** Effective January 1, 2026, CBAM now requires EU importers of cement, iron, steel, and aluminium to purchase certificates corresponding to the carbon price that would have been paid if goods were produced under EU carbon pricing rules. Transitional quarterly reporting on embedded emissions has been mandatory since October 2023. This directly impacts Kazakhstan exporters supplying these materials to the EU market. The cost is linked to the EU Emissions Trading System (ETS) allowance price, currently around 60-70 EUR per tonne of CO₂e.
- **Who it affects:** Kazakhstan exporters of steel, cement, and aluminium to the EU; EU importers; and ultimately, construction clients sourcing these materials via EU supply chains.
- **Effective date:** Financial obligations began January 1, 2026 (reporting mandatory since Oct 2023).
- **Business implication:** Increased material costs for carbon-intensive imports. Mandates granular supply chain transparency and verified emissions data. Drives demand for lower-carbon materials to reduce CBAM liability and maintain market access.
- **Source:** [CBAM for Construction — Guide 2026 | Plan Be Eco](<https://planbe.eco/en/blog/cbam-for-the-construction-industry>)
- **Kazakhstan Construction Code Effective July 1, 2026:** The new Construction Code of the Republic of Kazakhstan came into effect on July 1, 2026, replacing previous legislation. Key provisions include:
 - **Strengthened Control & Standardization:** Expands state architectural and construction control, and mandates the creation of a national institute for technical standardization.
 - **Digitalization:** Introduces a digital urban planning cadaster for ongoing monitoring, documentation registration, and record-keeping of commissioning certificates.
 - **Operational Control:** Implements operational control over facility technical condition post-commissioning to ensure safe operation.
 - **Extended Guarantees:** Mandates a minimum 10-year guarantee for supporting structures, roofs, and external walls/facades, and five years for other works.

- **Seismic Design:** Provides new provisions for design and construction in seismic zones.
- **Who it affects:** All construction companies, developers, contractors, and material manufacturers operating in Kazakhstan.
- **Effective date:** July 1, 2026 (certification of specialists from July 1, 2028).
- **Business implication:** Requires adherence to new technical standards and quality controls. Digitalization demands better data management. Longer guarantee periods increase contractor liability. Provides a framework for potentially integrating green building standards in the future.
- **Source:** [Kazakhstan Construction Code Goes into Effect](<https://www.morganlewis.com/pubs/2026/06/kazakhstan-construction-code-goes-into-effect>)

2. Construction & Market News

- **Kazakhstan Construction Materials Production Rises:** Kazakhstan's production of construction materials increased by 9.4% in January-July 2026, reaching 9.5 million tonnes. This surge aligns with a 15.2% rise in the value of construction work performed, which hit KZT 4 trillion in the first half of 2026. Despite growth, local production covers only about half of domestic demand, with significant reliance on imports for higher value-added segments like paints, HVAC, and sanitary ware. This indicates ongoing opportunities for local manufacturing investment, but also continued import exposure for certain product categories.
- **Source:** [Kazakhstan construction materials production grows in 2026](<https://economykz.org/en/kazakhstan-construction-materials-output-rises-9-4-amid-building-boom>)

3. ESG Disclosure Signals

Direct, recent ESG reports from major Kazakhstan construction companies for September 2026 were not readily available via general web search. However, the full implementation of CBAM acts as a de-facto mandatory disclosure requirement for EU importers sourcing materials from Kazakhstan, demanding verified embedded emissions data. The new Kazakhstan Construction Code's emphasis on digitalization of the urban planning cadaster (for monitoring and record-keeping) creates a foundational infrastructure that could facilitate future granular ESG reporting and compliance within the domestic market. The lack of public ESG reports for top Kazakh construction firms suggests a potential gap in transparency compared to global leaders, though the regulatory environment is beginning to build towards it.

4. Embodied Carbon Spotlight

Due to an ongoing tool error (lookup_carbon_factor and estimate_embodied_carbon are currently unavailable), a quantitative cradle-to-gate LCA could not be performed for this briefing. However, we can qualitatively assess a representative scenario and discuss industry trends.

Scenario: A typical concrete-intensive commercial building structure in Kazakhstan.

Carbon Hotspots: In such structures, cement (specifically clinker production) and steel (rebar, structural steel) are consistently the largest embodied carbon contributors. Cement manufacturing alone accounts for a significant percentage of global industrial CO₂ emissions.

Embodied Carbon Trends

Innovations in low-carbon concrete and circular economy approaches are rapidly advancing:

- **Biochar Integration:** Leading firms like Holcim are successfully embedding biochar from organic waste into cement, concrete, and mortars. This material actively sequesters carbon, with pilot projects demonstrating net-zero concrete with a footprint of -14 kg CO₂e/m³.

- **Scaling Calcined Clay:** Calcined clay is emerging as a critical supplementary cementitious material (SCM) that can halve the CO₂ footprint of cement without compromising performance, reducing dependence on traditional SCMs like fly ash and GGBS. Holcim projects 1 million tons of calcined clay cement production in 2026.
- **Advanced Concrete Recycling:** Proprietary crushing technologies allow for the recycling of used concrete back into aggregates and cement paste, closing the material loop and reducing reliance on virgin materials.
- **3D Concrete Printing:** This technology optimizes material use by building with less, accelerating construction, and facilitating the use of specialized, potentially lower-carbon mixes for specific structural elements.

These innovations offer significant pathways for reducing embodied carbon in construction projects, particularly in cement and concrete, which are directly impacted by CBAM.

Recommended Actions

1. **Immediate CBAM Supply Chain Audit:** For any materials (steel, cement, aluminium) exported to the EU, construction clients must immediately audit their supply chain. Identify direct and indirect importers, quantify volumes, and calculate potential CBAM liabilities based on EU ETS prices (currently €60-70/tCO₂e). Engage suppliers to secure verified actual production emissions data.
2. **Integrate CBAM Costs into Project Budgeting:** Update cost estimation and tendering models to factor in CBAM surcharges for imported materials. For long-term projects, include escalation clauses for fluctuating EU ETS carbon prices to avoid financial surprises.
3. **Prioritise Low-Carbon Material Procurement:** Actively seek out and specify low-carbon alternatives for concrete (e.g., using SCMs like calcined clay, GGBS, or biochar-infused concrete) and steel (e.g., Electric Arc Furnace (EAF) steel with high recycled content). These choices directly reduce embodied carbon and CBAM exposure.
4. **Leverage Kazakhstan's New Construction Code:** Understand the implications of the new national Construction Code on project planning, quality control, and long-term liabilities. Position the company to benefit from the new standardization and digitalization efforts, which could facilitate better EPD availability and green building practices in Kazakhstan.
5. **Develop Internal ESG Data Capabilities:** Given CBAM reporting requirements and the general trend towards greater transparency, invest in systems and expertise for collecting, verifying, and reporting environmental data (especially embodied carbon) at a granular material level.

Sources

- [CBAM for Construction — Guide 2026 | Plan Be Eco](<https://planbe.eco/en/blog/cbam-for-the-construction-industry>)
- [Kazakhstan Construction Code Goes into Effect](<https://www.morganlewis.com/pubs/2026/06/kazakhstan-construction-code-goes-into-effect>)
- [Kazakhstan construction materials production grows in 2026](<https://economykz.org/en/kazakhstan-construction-materials-output-rises-9-4-amid-building-boom>)
- [5 construction innovations to watch in 2026](<https://www.holcim.com/who-we-are/our-stories/construction-innovations-in-2026>)

Generated by the ESG Intelligence Agent on September 26, 2026. Embodied-carbon figures are indicative cradle-to-gate (A1-A3) estimates — verify with product-specific EPDs before formal reporting. Specific embodied carbon calculations could not be performed this run due to a tool error.